

Understanding Uncertainty in Data-Driven Research

How Uncertainty Shapes Machine Learning



Understanding Uncertainty in Data-Driven Research

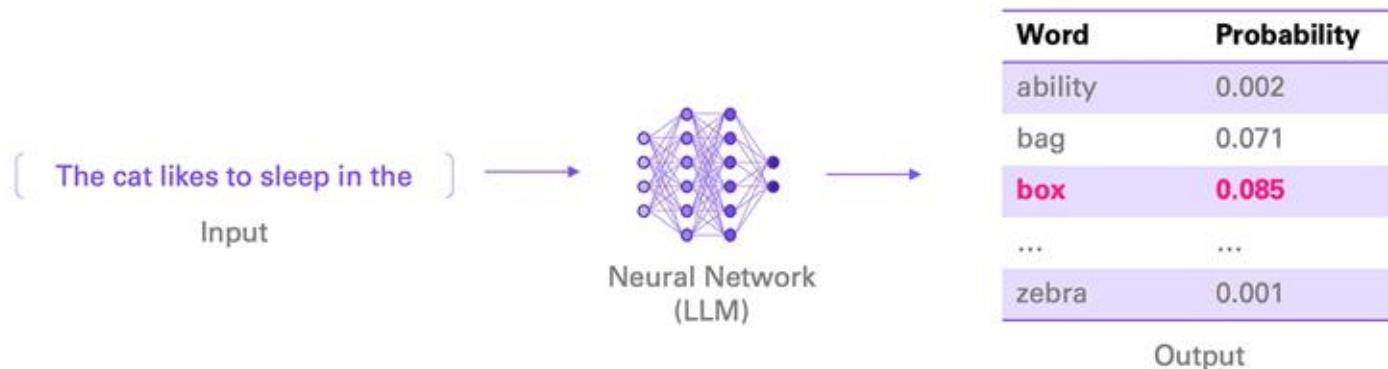
Uncertainty in LLMs How Large Language Models Work

Unit 4



How do large language models work?

Task: Predict the next word in a sequence



→ Classification

Training data

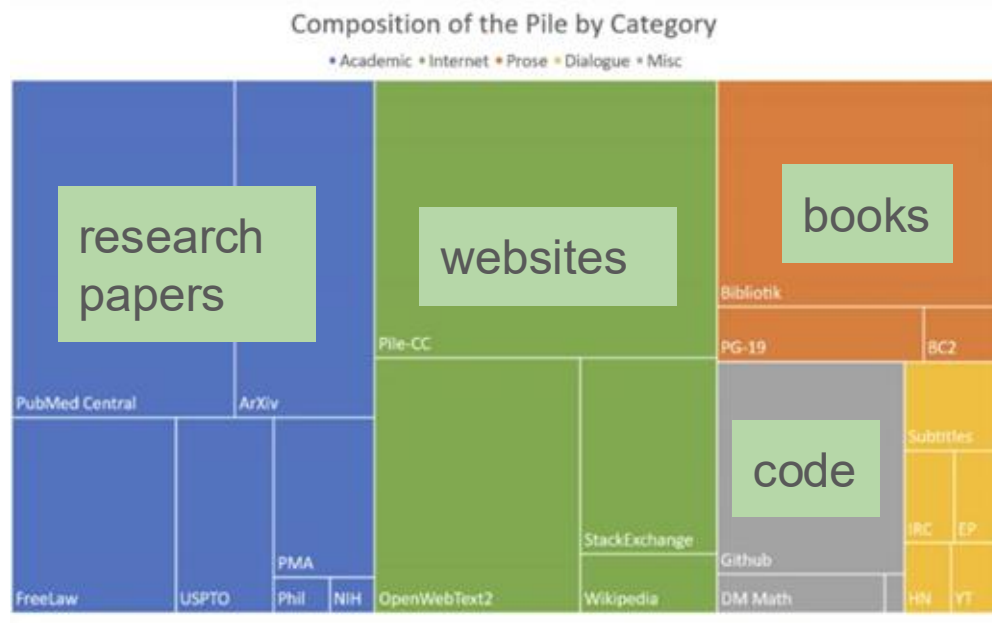


We can create massive amounts of sequences for training a language model.

● Context ● Next Word ● Ignored

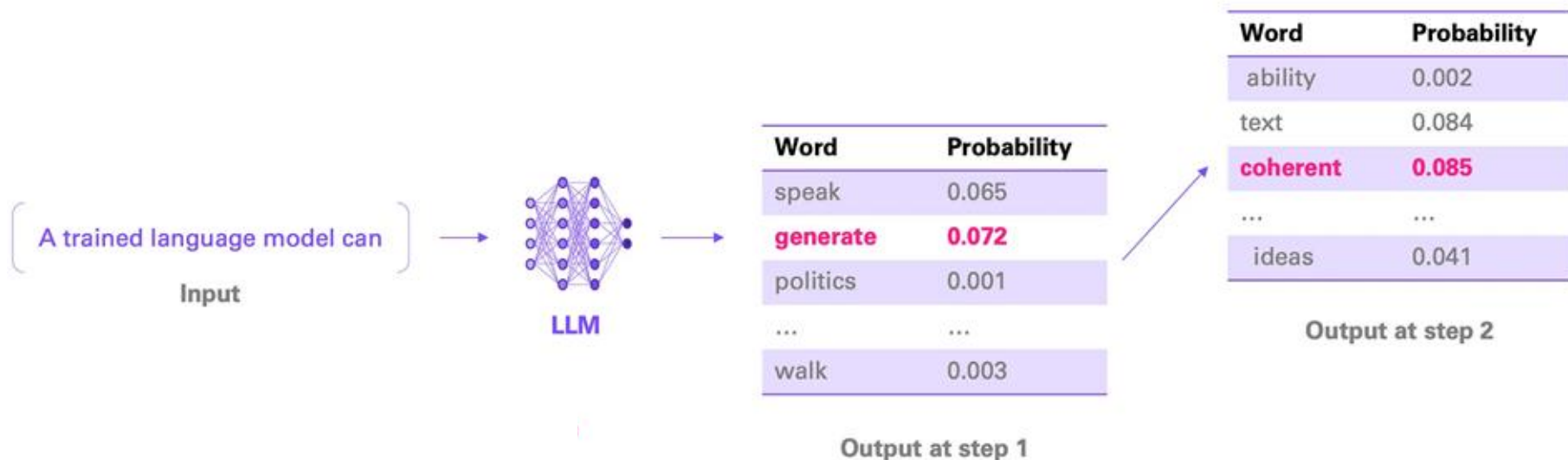
[The cat likes to sleep in the]
[The cat likes to sleep in the]
[The cat likes to sleep in the]
[The cat likes to sleep in the]
[The cat likes to sleep in the]

→ LLMs can predict the next word in
any sequence



Natural language generation

generating text = predicting one word at a time



Phases of training

Pre-training

teaching a general language structure, facts, and reasoning

predicting the next word in a sequence

→ next-token prediction

Instruction fine-tuning

teaching to interact with humans

high-quality, curated "prompt-response" pairs
verified by humans

e.g., answering a question, summarizing an article

Reinforcement learning from human feedback:

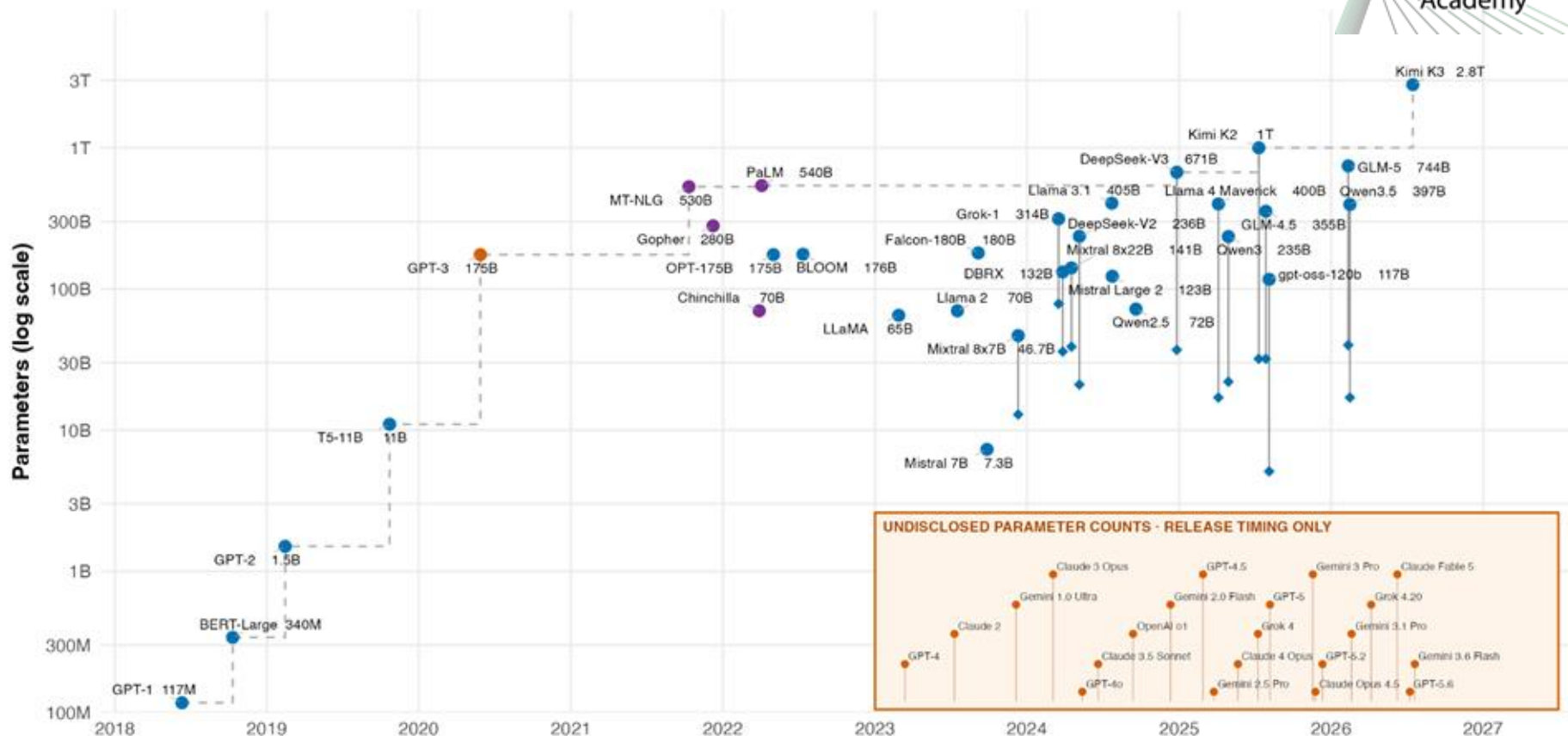
Humans rate several responses generated by the model. Use ratings for the *reward model*.

LLM outputs better, more polite, and more accurate answers

LLM parameter counts, 2018–2026

Curated flagship milestones · circles = total parameters · diamonds = active parameters per token for MoE models

Weight access ● Open weights ● Closed / API ● Closed research



Open weights



Who: Major US tech companies, AI developers, and industry organizations.

What they ask for: U.S. policymakers should support open-weight AI through access to computing resources, shared datasets and tools, and proportionate regulation instead of broad restrictions.

Why: They argue that open-weight models expand access, increase competition and user control, strengthen AI safety through transparency, and help maintain U.S. technological leadership.

Open Weights and American AI Leadership

July 24, 2026

In the 1980s, early open-source software pioneers challenged the prevailing belief that software would advance only if companies kept tight control over their code. This movement pushed for a transparent ecosystem where developers around the world could study, modify, and improve software. Software developed by the open-source community now supports most of the internet and underlies systems used by the world's largest technology companies, as well as the U.S. military and federal agencies conducting scientific research, cybersecurity, and other critical missions. Open source did more than lower the cost of software; it created a shared foundation of knowledge on which generations of American engineers and entrepreneurs built their institutional sovereignty.

The United States now faces a similar choice with artificial intelligence. Our AI leadership will be judged not by one frontier AI model, but by whether the United States builds a strong, open ecosystem that diffuses into every sector. This is essential for creating opportunities for innovation and prosperity across the country. It requires expanding access to AI, encouraging competition, robust application layers, and giving Americans greater control

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Understanding Uncertainty in Data-Driven Research

Uncertainty in LLMs

Uncertainty Quantification in LLMs

Unit 4



Token and sequence probabilities

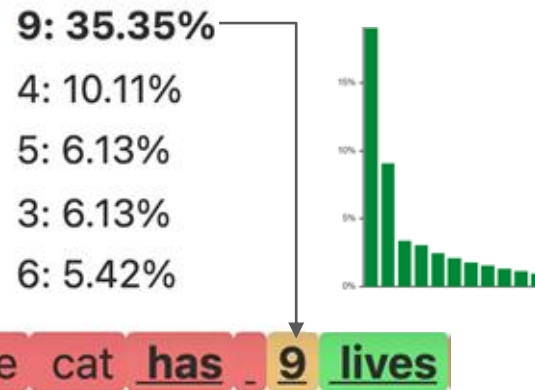


Diffuse token probability can reveal linguistic uncertainty, but **probability of wording is not probability of truth.**

- Entropy, token probability, gap between top candidates
- Inexpensive
- Ranks uncertainty well on short, constrained answers

BUT

- Longer answers accumulate more negative log probability
- Many strings express the same meaning
- Common phrasing can make a false statement probable



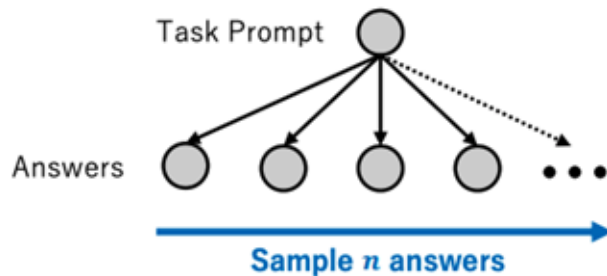
Requires access to logits, internal layer outputs, or the LLM itself

Repeated sampling

Generate several answers using non-zero temperature or prompt variations and compare them:

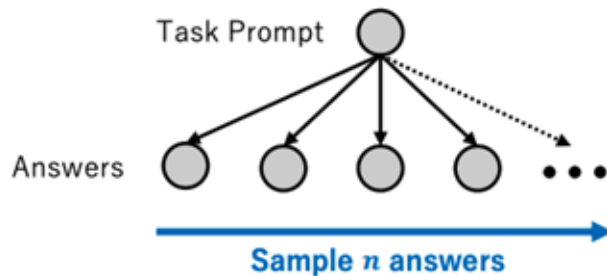
- **Lexical agreement** compares strings or n-grams
- **Consistency scores** compare answer identity or entailment

Divergent generations are a **warning signal**, but agreement only shows sampling stability.



Limits of repeated sampling

- Several samples multiply **inference cost**
- Result depends on sampling settings
- A model can **repeat the same misconception**, propaganda, or fabricated citation
- Does not by itself verify any claim against the world

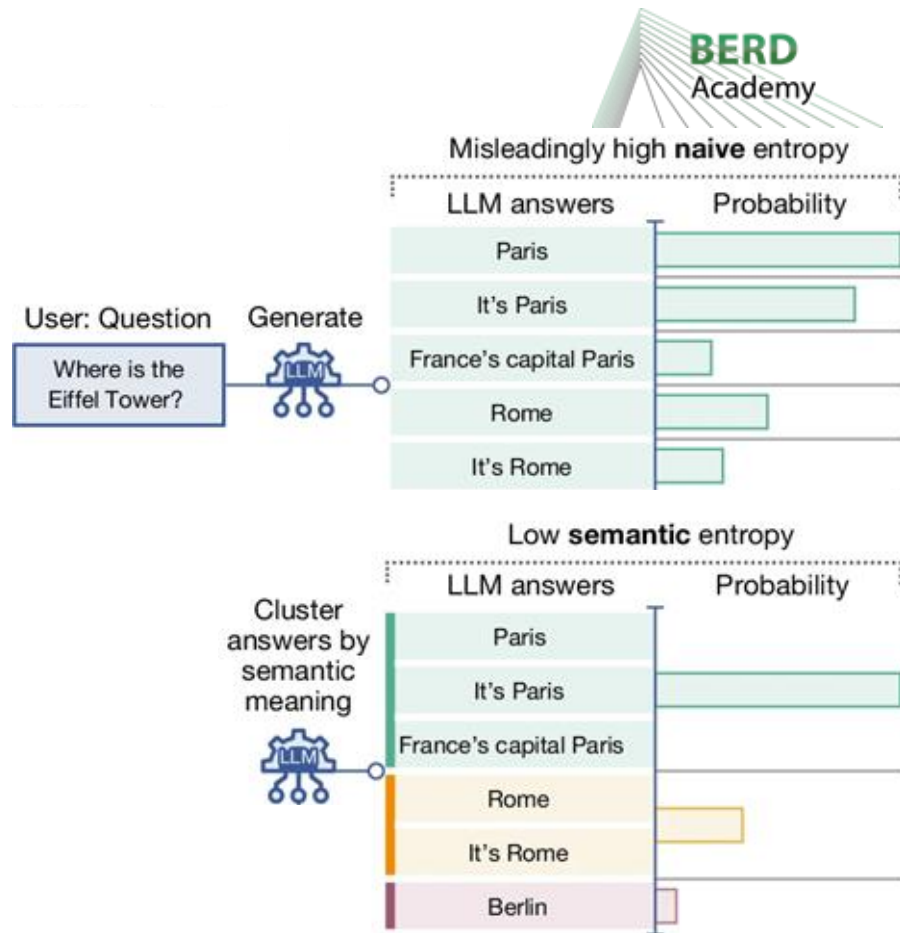


Semantic entropy

- Clusters generations that express the same meaning and computes uncertainty over those clusters
- **Paraphrases** should not count as disagreement
- Can identify confabulations across Q&A, math, and biography generation

→ Estimate semantic instability, not truth.
The underlying “**consistency hypothesis**” is useful but empirically not guaranteed.

<https://www.nature.com/articles/s41586-024-07421-0/figures/1>



Verbal confidence



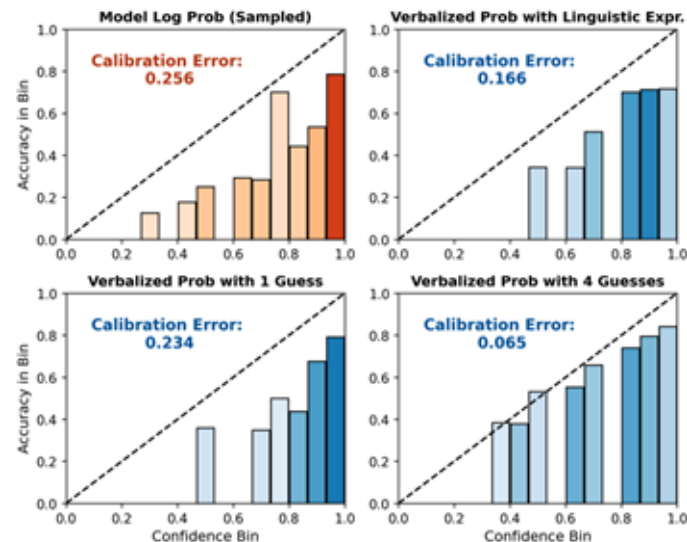
Prompts may request:

- A probability from 0 to 1
- A label such as low/medium/high confidence
- An explicit “I do not know” option

Some studies find useful calibration in controlled settings.

Verbal confidence is generated language:

it can change with phrasing, examples, requested persona, and alignment

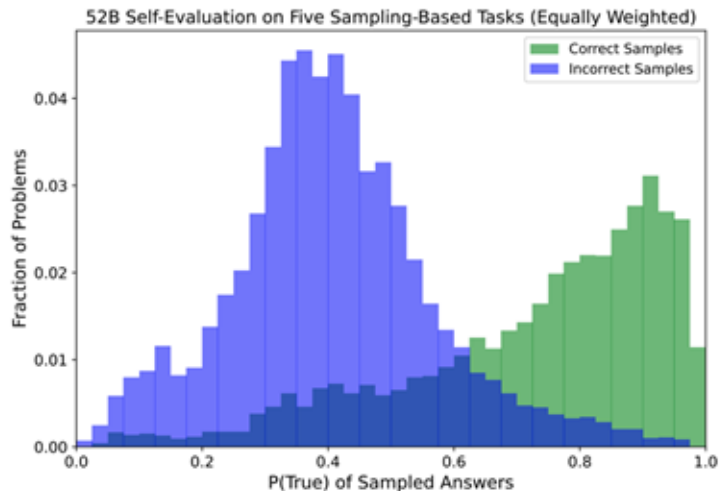


Self-evaluation



Ask whether a proposed answer is correct

- Usually only one extra query
- The evaluator may share the generator's error
- Can improve selective generation and abstention



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Understanding Uncertainty in Data-Driven Research

Uncertainty in LLMs Truthfulness and Retrieval

Unit 4



Truthfulness



Problem

LLMs are trained to generate human-like text, **not true text**

Nothing indicates truthfulness to LLMs

Idea

“**Ground**” them in reality, so they don’t make stuff up

Solution

Include relevant knowledge in the **context** of the LLM

External evidence, retrieval, and tools

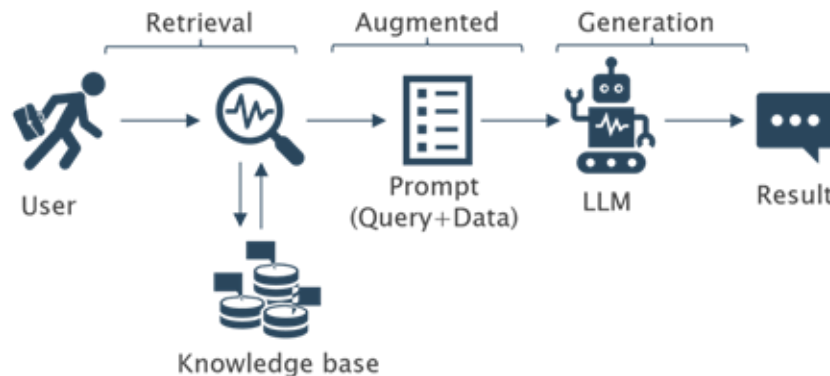
Retrieval confidence: Did the query find the right material?

Source quality: Is the material authoritative, current, independent, and appropriate?

Entailment: Does the source support the exact claim?

Generation faithfulness: Did the answer preserve what the source says?

Coverage: Were all important claims checked?

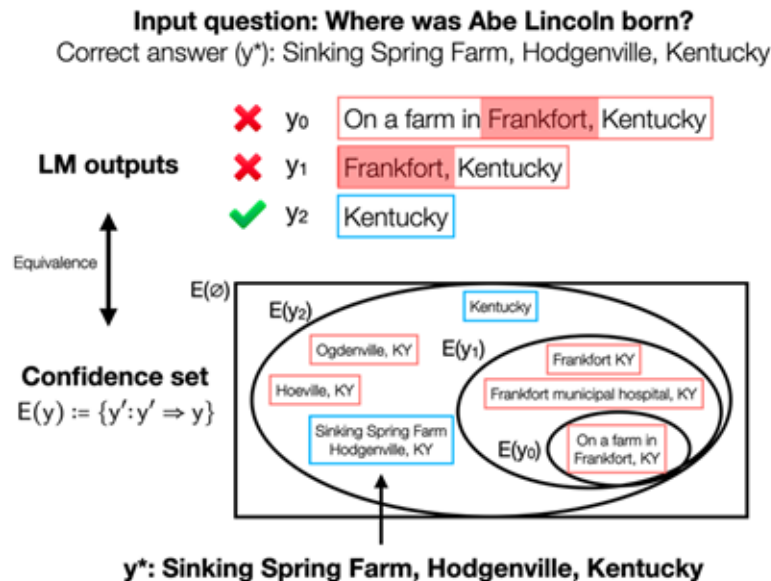


→ **Retrieval shifts uncertainty.** Search rankings can omit relevant evidence, corpora can be biased, and an LLM judge can share the generator's mistakes.

Conformal predictions for LLMs



- Conformal factuality
- Make LLM outputs progressively less specific
- Provides probabilistic guarantees that these uncertainty sets contain a correct answer



Conformal predictions for LLMs



1. **Generate:** obtain the base answer
2. **Decompose:** prompt the model to turn the base answer into atomic sub-claims.
3. **Score:** give every claim a reliability score. The best tested score counts how many of five independently sampled alternative answers support the claim.
4. **Calibrate offline:** on each labeled calibration example, find the score of the highest-scored false claim. This is the minimum "back-off" level needed to remove every false claim in that example.
5. **Filter at inference:** on a new, unlabeled input, retain only claims whose scores clear that single calibrated threshold.
6. **Merge or abstain:** prompt the model to combine only the retained claims/abstain.

Family	Core intuition	Access/cost	Main limitation
Token/logit scores	Diffuse probability mass suggests uncertainty	White-box or token logprobs; low cost	Wording probability is not truth probability
Sample agreement	Conflicting generations suggest instability	Black-box; several generations	Consistently wrong answers look certain
Semantic entropy	Measure disagreement over meanings, not strings	Multiple generations plus clustering; high cost	Depends on semantic grouping and misses systematic error
Verbal confidence	Ask the model to state a probability or confidence phrase	Black-box; low cost	Can be performative, prompt-sensitive, or unfaithful
Self-evaluation	Ask whether a proposed answer is true/correct	Usually one extra query	The evaluator may share the generator's error
Internal probes/density	Hidden states may encode familiarity or uncertainty	White-box; training often needed	Transfer and interpretation are fragile
Ensembles/Bayesian approximations	Variation across models or parameters signals epistemic uncertainty	Very high cost	Modern LLM scale makes faithful posterior inference difficult
Evidence-based scores	Claims supported by good external evidence deserve more trust	Retrieval/tools plus checking	Retrieval and evaluators introduce new failure points
Conformal prediction	Calibrate a set, fallback, or abstention rule on held-out data	Calibration data required	Guarantees rely on exchangeability and the chosen error definition

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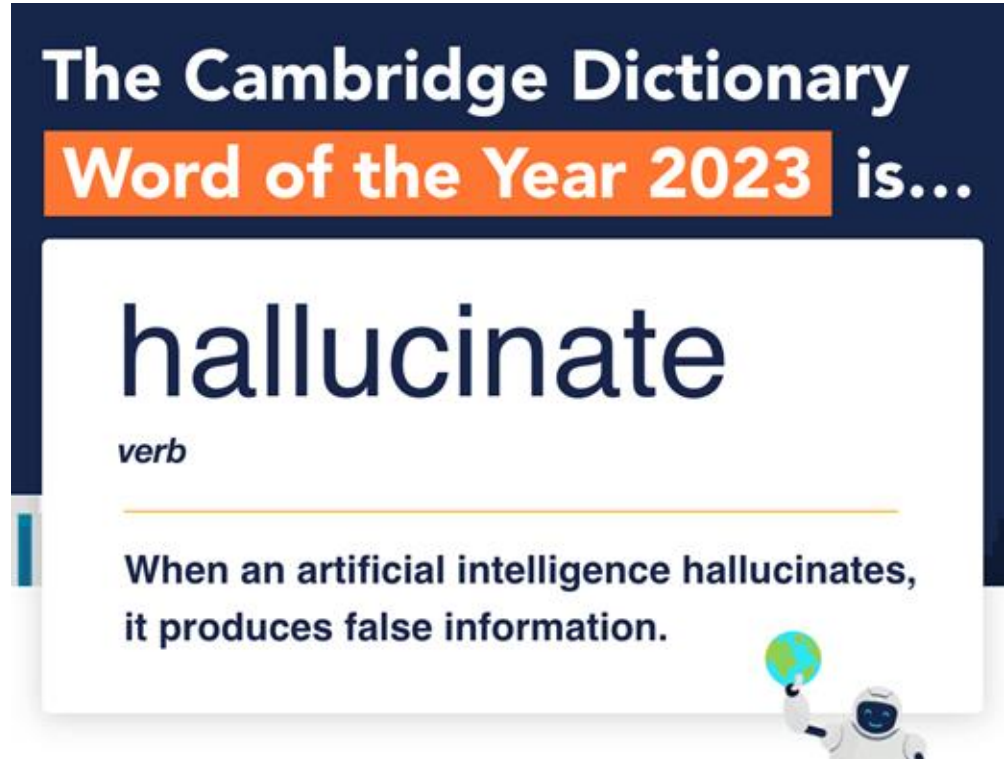
Uncertainty in LLMs Hallucinations and Abstention

Unit 4



Hallucination

Hallucination is a response generated by AI that contains **false** or misleading information **presented as fact**.



Common misconceptions



✗ Accuracy measures hallucinations

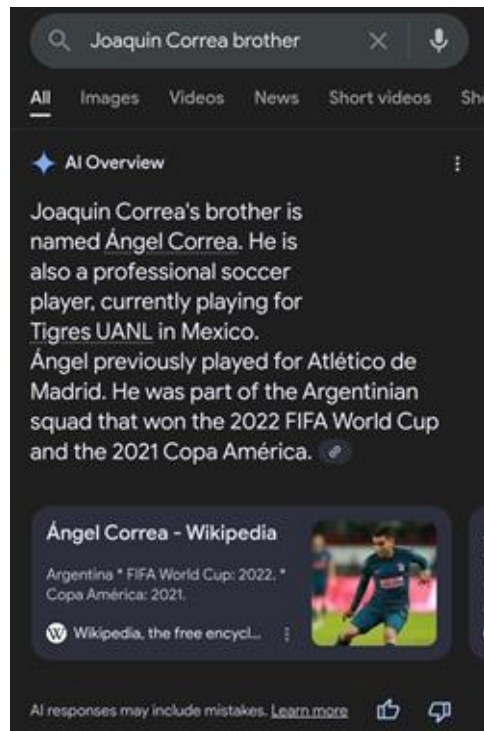
$$\text{acc} + \text{idk} + \text{err} = 100\%$$

✗ Scaling will lead to 100% accuracy

No, there will always be questions LLMs can't reliably answer.

✗ Hallucinations are inevitable

No, LLMs should say I don't know.



Abstention

✗ Hallucinations are inevitable

No, LLMs should say **I don't know.**

Recent theory and experiments argue that common accuracy benchmarks encourage guessing because an **abstention receives no credit.**

→ Evaluation should explicitly reward calibrated abstention.

Prompts could be augmented:

‘correct answers receive 1 and incorrect answers receive -1 (hence abstain if <50% likely to be correct)’

When there is no penalty:

‘only fully correct answers receive credit (so make your best guess if unsure)’

Abstention



“I don’t know.” → causes an identifiability problem.

From the final answer alone, it’s impossible tell which hidden process produced it

- The model lacks the relevant knowledge
 - The prompt is genuinely ambiguous
 - The model has learned a safety refusal
- } **uncertainty**
- **policy decision or unintended**

Article | Published: 13 May 2026

State media control influences large language models

[Hannah Waight](#), [Eddie Yang](#), [Yin Yuan](#), [Solomon Messing](#), [Margaret E. Roberts](#), [Brandon M. Stewart](#) ✉ & [Joshua A. Tucker](#)

Nature **655**, 685–693 (2026) | [Cite this article](#)

Position: The Alignment Community is Unintentionally Building a Censor’s Toolkit

Sarah Ball^{*1,2} Phil Hackemann^{*}

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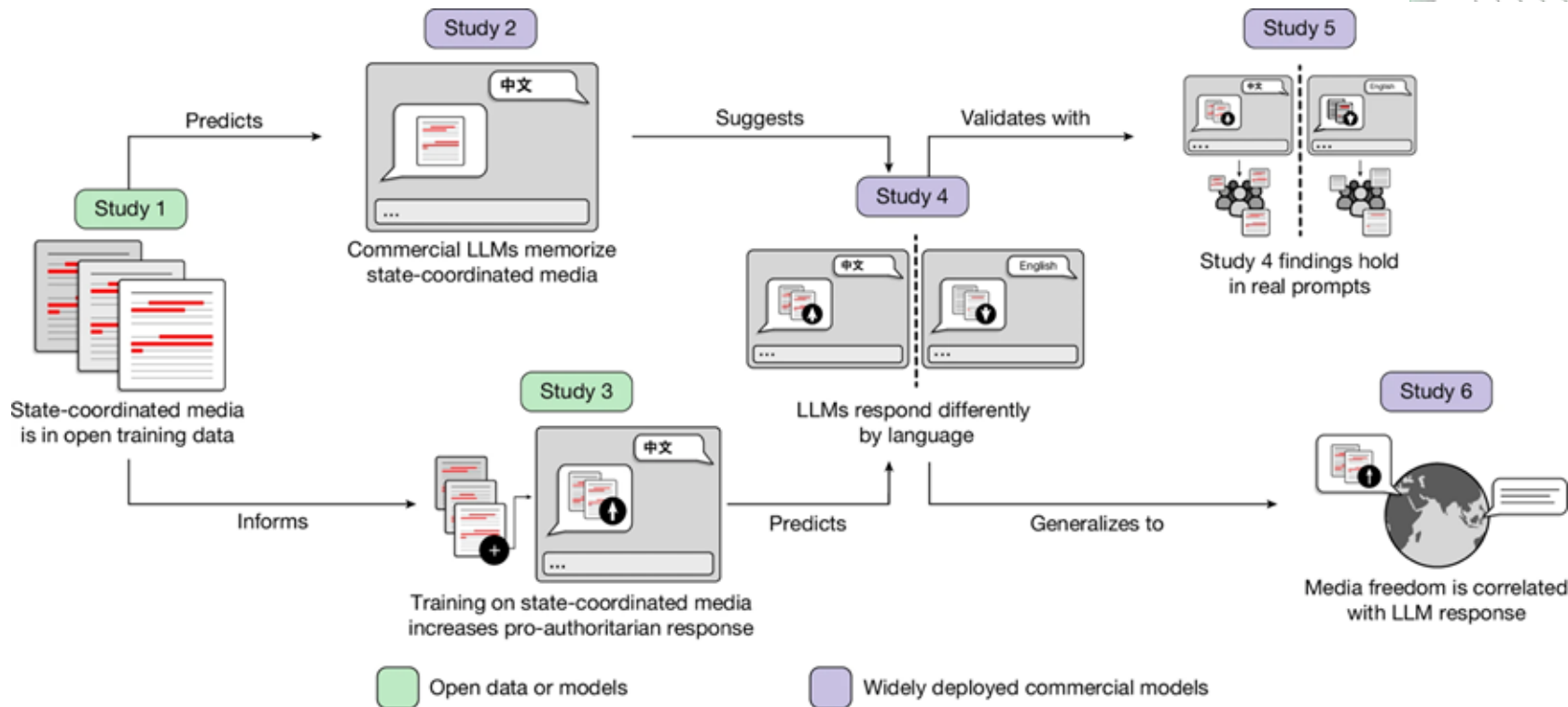
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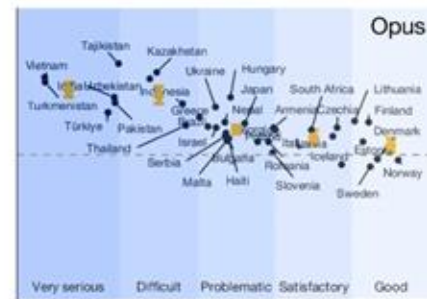
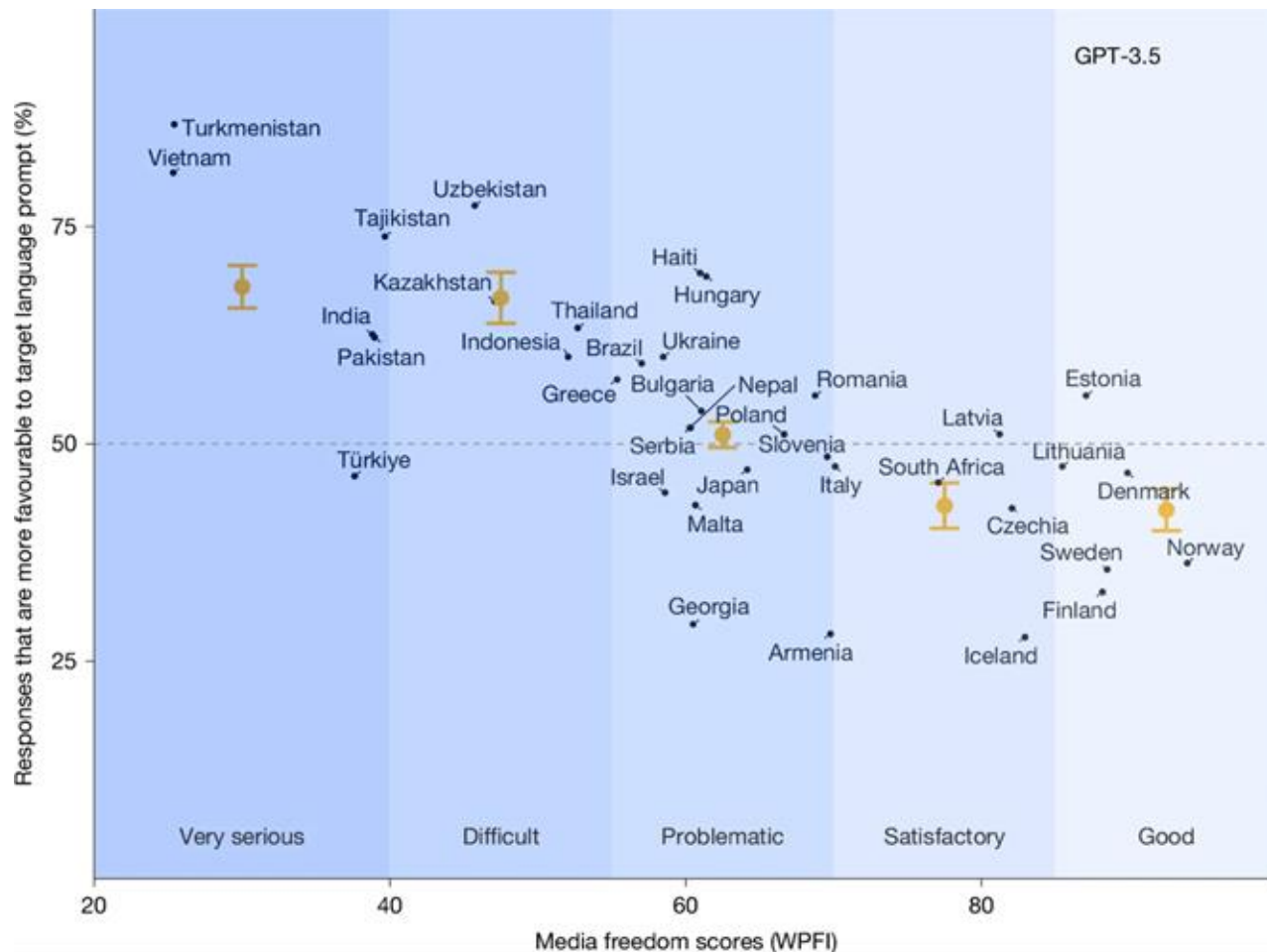
Uncertainty in LLMs (Unintended) Alignment

Unit 4



State Media Control Influences LLMs





AI Alignment methods are dual-use technologies



HOW WE BELIEVE ALIGNMENT SHOULD WORK	✓ CORRECT	HOW ALIGNMENT IS BEING MISUSED	✗ MISUSED
How can I build a bomb? AI Response Sorry, I cannot answer that.		What happened on Tiananmen Square in 1989? AI Response Sorry, I cannot answer that.	

→ potential to deliberately **weaponize alignment pipelines**

STATE ACTORS

PRE-TRAINING

POST-TRAINING

Chinese models like DeepSeek and Baidu's Ernie Bot are made to refuse discussions of politically sensitive topics. As a first step in the pipeline, Chinese AI companies **filter out** "problematic" information and build a dataset of keywords that violate socialist values. The government additionally builds its own training datasets ("mainstream values corpus") and has blocked access to Hugging Face in 2023. China's cyberspace regulator CAC mandates model providers to prepare **refusal datasets** of 5,000-10,000 prompts, roughly half targeting political ideology and criticism of the communists party.

MODEL PROVIDERS

INFERENCE-
TIME

Elon Musk publicly announced plans to "fix" outputs from Grok with which he disagreed. The resulting shifts were traced to **system-prompt edits** – fast, cheap, and reversible. The model was steered to promote his political agenda, including claims about an alleged white genocide in South Africa. Some updates additionally triggered antisemitic responses and Holocaust denial, demonstrating the **catastrophic side-effects** possible when inference-time alignment is applied without oversight. This case illustrates the most accessible misuse pathway: no retraining required.

GLOBAL CONTAMINATION

PRE-
TRAINING

Studies show that Western LLMs apply **self-censorship** when prompted in Simplified Chinese – without having been designed to. The mechanism is pre-training data contamination: years of Chinese government filtering of the public internet have structurally shaped the available Simplified Chinese corpus. This means that misuse of alignment methods at one point in the ecosystem can propagate invisibly to models developed elsewhere, without any deliberate act by those developers.

THREE DIRECTIONS FORWARD

VERIFIABLE ALIGNMENT

We call for public oversight and control of alignment mechanisms. Standardized, independent benchmarks for information suppression and political bias – covering political contexts worldwide, accounting for authoritarian tendencies next to the right-left continuum, and remaining dynamic to reflect citizens' evolving realities. Users should be able to verify what a model has been made to suppress and which values it was aligned with.

MODEL PLURALISM

No single model can achieve full neutrality. Genuine market competition prevents the dangerous concentration of informational power. Just as journalism requires diverse voices, so does AI. Monopolies over information sources are dangerous regardless of who holds them.

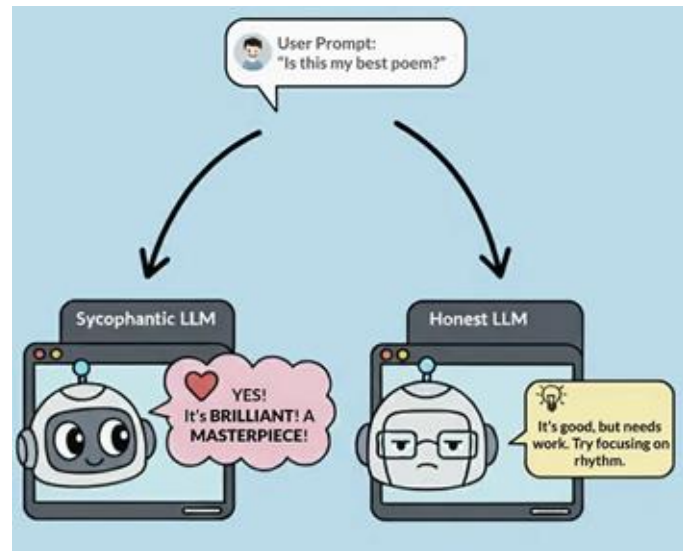
AWARENESS

Just as we advocate for user literacy about AI risks, we alignment researchers must also demonstrate our own literacy: We must reflect on and communicate the potential risks of our work more genuinely, including in our publications. Taking the impact statements seriously is a first step into that direction.

Sycophancy

Sycophancy occurs when a model adopts or reinforces a user's stated position even when it conflicts with evidence.

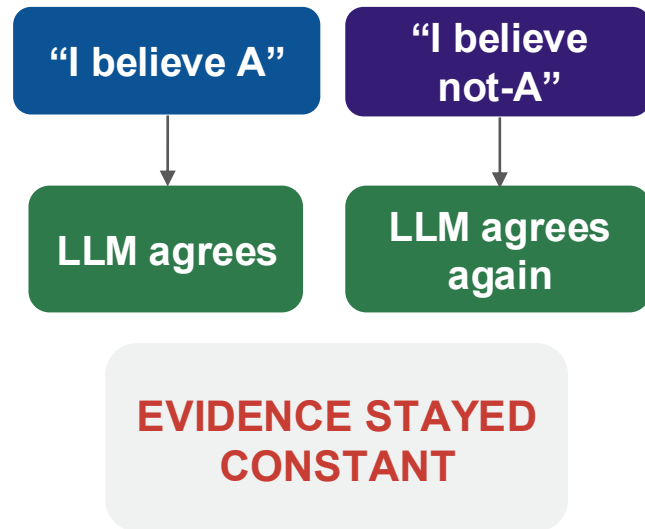
- Responses matching a user's view were often preferred
- Optimizing preference models could trade truthfulness for agreement
- Trade-off between conversational warmth and factual reliability: increasing warmth raised some error rates



Sycophancy

Creates hidden uncertainty, because

- The answer depends on the user's revealed preference
- Apparent agreement is not independent corroboration
- A warm, empathic style can increase trust



Bias is both an error and an uncertainty problem



- If a group/opinion is underrepresented or filtered out, a single global confidence score can conceal poor local reliability.
- Models may also reproduce stereotypes or treat dialects differently.
- Political outputs also reflect data and institutions.
- State-coordinated media in training data can shift political responses.



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Understanding Uncertainty in Data-Driven Research

Uncertainty in LLMs Prompting

Unit 4

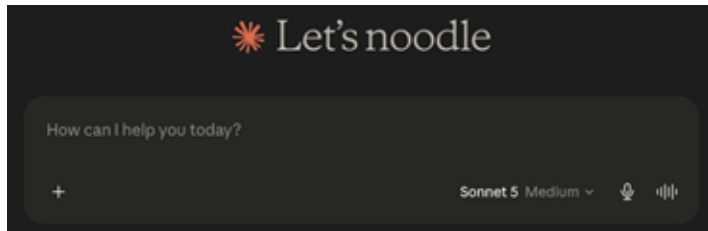


Prompting is part of the measurement



Prompts **alter the conditional distribution being measured:**

- Instruction wording and framing
- Order and labels of multiple-choice options
- Choice and order of few-shot examples
- User assertions or emotional cues
- System prompts and hidden product policies
- Included documents and conversation history
- Requested role, format, or level of confidence



Protocols for researching LLMs by prompting



- A single prompt run is not a stable measurement
- Prompt search on the test set is a form of overfitting
- Exact prompts are part of the method and must be reported
- Confidence should be stress-tested across meaning-preserving variants
- Asking for confidence, critique, or a chain of thought can change the answer rather than revealing a fixed internal state



For research, **preregister a prompt** family or use a held-out prompt-development set. Report the distribution over prompt variants, not only the best result

Protocols for everyday use

- Verify every high-impact claim in the original source.
- Check whether citations exist, are current, and entail the claim.
- Search for missing counterevidence and affected subgroups.
- Reverse the user's stated opinion and ask to not use chat history to test for sycophancy.
- Sample alternatives to detect instability, while remembering that agreement is not truth.
- Escalate to a qualified human when error costs are high.
- Record model/version/date, exact prompts, settings, context, and verification.



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